

Heesu Kim

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EDUCATION

Sungkyunkwan University

BEng, Applied Artificial Intelligence GPA: 4.05/4.5

Seoul, South Korea

Feb 2021 – Aug 2027

Hitotsubashi University

Exchange Student

Tokyo, Japan

Sep 2025 – Aug 2026

EXPERIENCE

Undergraduate Research Intern

Komachi Lab, Hitotsubashi University

Sep 2025 – Aug 2026

- Worked on creating a dataset for measuring cultural bias and took part in the creation of the Korean subset
- Participated in multiple projects related to semantic shift detection

Undergraduate Research Intern

MARS Lab, Samsung Medical Center

Jul 2025 – Aug 2025

- Assigned to object detection and segmentation tasks in medical domain
- Fine-tuned foundational models for detection and segmentation of post-surgery skin flaps

Army Sergeant

Defense Security Agency, Ministry of Defense

Apr 2023 – Oct 2024

- Assisted with US-ROK joint operation through translation and interpretation
- Created and managed tools that automated most of the workflow during service

PUBLICATIONS

Retrieval Without Relevance: Prompt Dependence in Small Machine Translation Models

Heesu Kim, Shota Sato, Mamoru Komachi

Oct 2026

BabyLM Workshop

- Trained SLMs from scratch under strict rule set to measure how RAG affects machine translation
- Compared the machine translation performance across several conditions such as SFT and RAG

Hi-DARTS: Hierarchical Dynamically Adapting Reinforcement Trading System

Hoon Sagong, Heesu Kim, Hanbeen Hong

Oct 2025

ICTC 2025

- Developed a multi-layer trading architecture based on reinforcement learning algorithms
- Benchmarked our agent against static strategies such as buy and hold
- Showed efficient selection of sub-agents based on volatility of the selected equity

PROJECTS

Qualitative Analysis of Agentic Traces Across Quantization Levels | *llama.cpp, vLLM*

Aug 2026 –

- Performed both qualitative and quantitative evaluation across quantization levels for the Qwen 3.6 27b model
- Used various coding benchmarks: MBPP+, LiveCodeBench, BFCL v4, SWE-bench lite
- Applying LLM-as-a-judge to qualitatively review agentic traces of quantized models

Closed-set Object Detection & Segmentation | *Python, PyTorch*

Jul 2025 – Aug 2025

- Constructed a pipeline for fine-tuning and benchmarking different detection architectures
- Used YOLOv8, YOLOv11, RF-DETR as the base model for fine-tuning
- Created a custom single-class, monolithic, semantic segmentation model with DINOv3 as the backbone

Workflow Process Automation Tool | *PowerShell*

Dec 2023 – Jun 2024

- Implemented basic macro functions by accessing Win32 API through PowerShell
- Analyzed the assigned workflow and found repetitive actions that can be automated
- Entire development process occurred in an area with restricted internet access

Compression Utility Using Huffman Code | *C*

Nov 2022 – Dec 2022

- Implemented Huffman Code in C
- Created a compression CLI tool based on the implementation
- Achieved a compression ratio of 1.8 on selected text files

AWARDS

3rd Place, Bias-a-Thon 2025 Track 1

Jun 2025

Awarded by the Dean of the College of Computing and Informatics, Sungkyunkwan University

- Recognized for identifying and addressing biases against certain demographics in commonly used Large Language Models
- Hackathon hosted by the Center for Multimedia Intelligence, Sungkyunkwan University

TECHNICAL SKILLS

Languages: Python, C, JavaScript, R, PowerShell

Developer Tools: Git, VS Code, Visual Studio

Libraries: PyTorch, pandas, NumPy, Matplotlib, Seaborn, vLLM